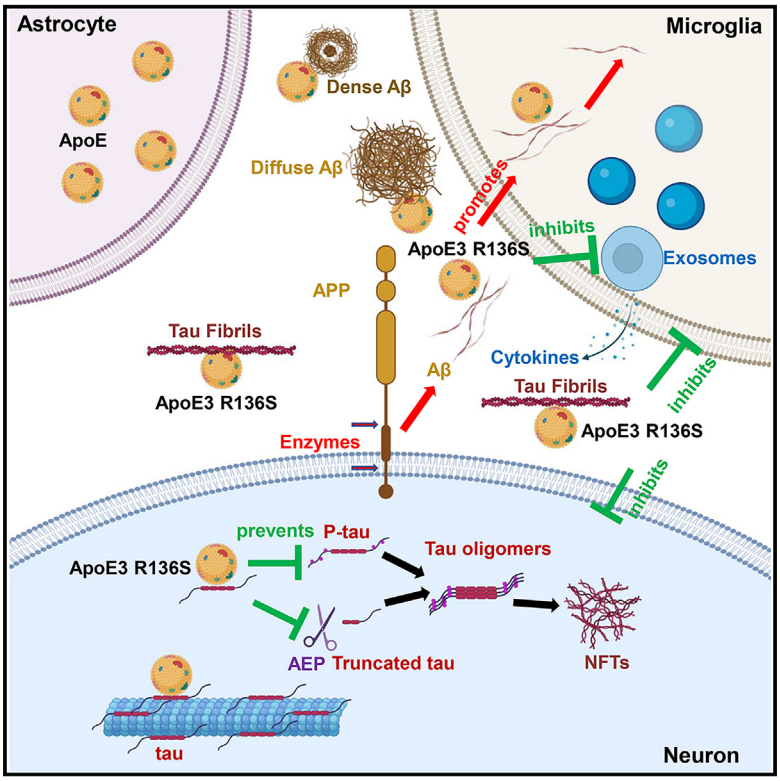
Neuron

Article

ApoE3 R136S binds to Tau and blocks its propagation, suppressing neurodegeneration in mice with Alzheimer's disease

Graphical abstract



Highlights

- ApoE3 R136S strongly binds to Tau and reduces its uptake into neurons and microglia
- ApoE3 R136S diminishes Tau fragmentation by AEP and reduces neurotoxicity
- ApoE3 R136S attenuates Tau spreading in the brains of both 5xFAD and Tau P301S mice

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In brief

Guiqin Chen, Mengmeng Wang, et al. demonstrate that ApoE3 R136S inhibits AEP-mediated Tau cleavage and neuroinflammation by binding more tightly to Tau, preventing its propagation, and thereby slowing the progression of Alzheimer's disease in mice.

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ApoE3 R136S binds to Tau and blocks its propagation, suppressing neurodegeneration in mice with Alzheimer's disease

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SUMMARY

PSEN1 E280A carrier for the APOE3 Christchurch variant (R136S) is protected against Alzheimer's disease (AD) symptoms with a distinct anatomical pattern of Tau pathology. However, the molecular mechanism accounting for this protective effect remains incompletely understood. Here, we show that the ApoE3 R136S mutant strongly binds to Tau and reduces its uptake into neurons and microglia compared with ApoE3 wild type (WT), diminishing Tau fragmentation by asparagine endopeptidase (AEP), proinflammatory cytokines by Tau pre-formed fibrils (PFFs) or β-amyloid (Aβ), and neurotoxicity. Further, ApoE3 R136S demonstrates more robust effects in attenuating AEP activation and Tau PFF spreading in the brains of both 5xFAD and Tau P301S mice than in ApoE3 WT, leading to improved cognitive functions. Thus, our findings support the idea that ApoE3 R136S strongly binds Tau and decreases its cellular uptake, abrogating Tau pathology propagation in AD brains.

INTRODUCTION

Alzheimer's disease (AD) is the most common neurodegenerative disease, characterized by the extracellular senile plaques, mainly consisting of aggregated β-amyloid (Aβ) polypeptide, and intraneuronal neurofibrillary tangles (NFT) predominantly composed of truncated and hyperphosphorylated Tau. Familial AD (FAD) is a very rare autosomal dominant disease with early onset (~45 years old), caused by mutations in amyloid precursor protein (APP), PSEN1, or PSEN2 that lead to aberrant Aβ overproduction and aggregation. By contrast, sporadic AD (SAD) is the vast majority.[1]

Apolipoprotein E4 (ApoE4), the most common genetic risk factor for AD, directly influences disease onset and progression by impacting the major pathological hallmarks of AD, including Aβ plaques, NFTs, and neuroinflammation.[2] The three APOE alleles encode protein isoforms that differ from one another only at amino acid positions 112 and 158: ApoE2 (C112, C158), ApoE3 (C112, R158), and ApoE4 (R112, R158). It is becoming increasingly clear that the biochemical and biophysical properties of ApoE trigger a cascade of events at the cellular and systems levels, ultimately impacting aging-related pathogenic conditions, including AD.[3] ApoE is a major protein of the lipoprotein transport system that plays important roles in lipid homeostasis and protection from atherosclerosis. Mutations in the 136–150 region of the N-terminal domain of ApoE reduce its low-density lipoprotein (LDL) receptor binding capacity and have been linked with lipoprotein disorders, such as type III hyperlipoproteinemia (HLP) in humans.[4]

Recently, it was reported that a *PSEN1* E280A mutation carrier from the world's largest known autosomal dominant AD kindred who was spared from Alzheimer's symptoms until her seventies, nearly three decades after the typical age of clinical onset among mutation carriers in these kindred. The old patient carries two copies of the rare ApoE3 Christchurch variant (APOE3ch, R136S) and displays severely elevated brain Aβ compared with others in the kindred with limited Tau pathology, as measured by *in vivo* positron emission tomography (PET) imaging.[5] Longitudinal Tau PET study reveals a distinct anatomical pattern of Tau pathology with atypical accumulation *in vivo* and unusual postmortem regional distribution characterized by sparing in the frontal cortex and severe pathology in the

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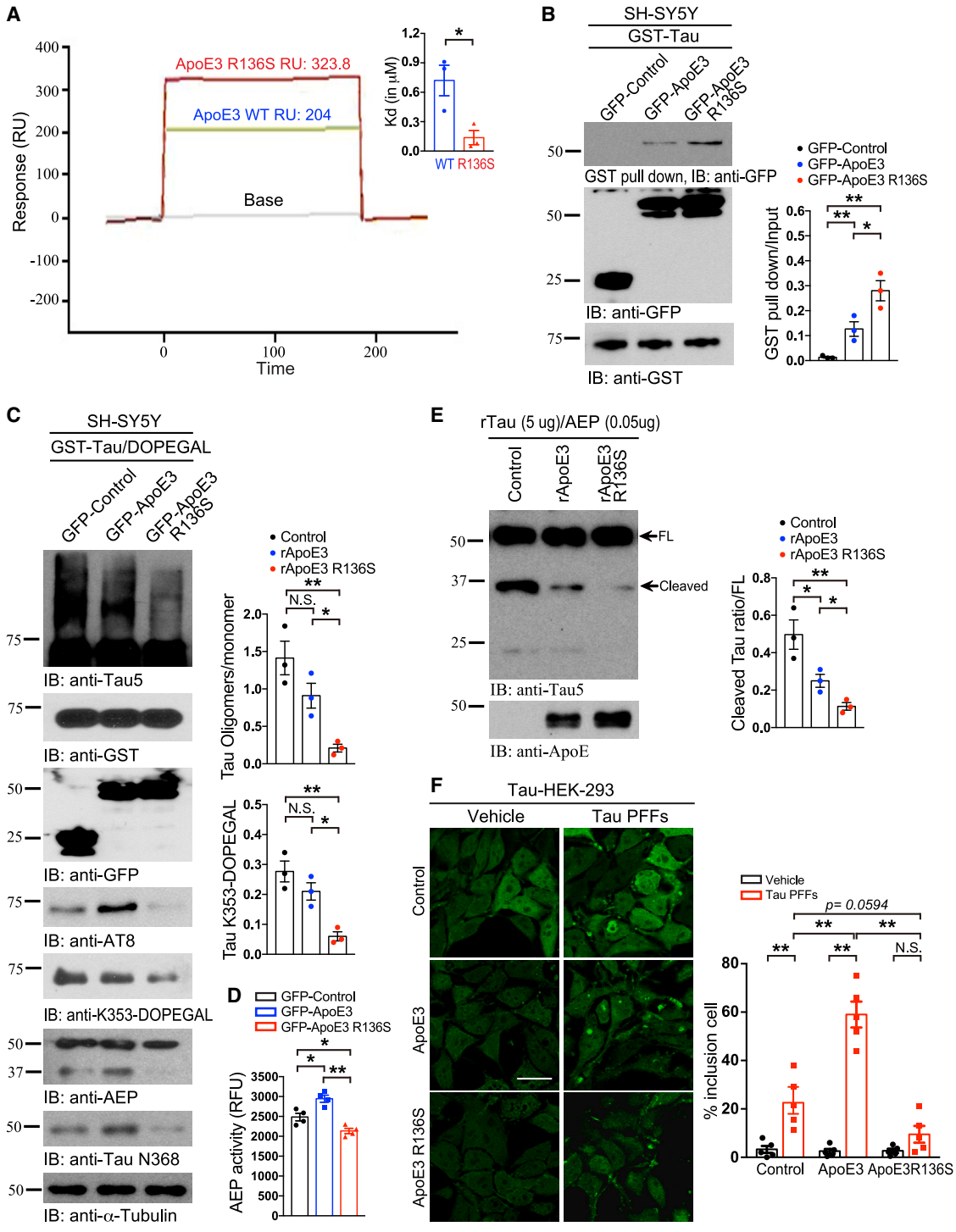


Figure 1. ApoE3 R136S displays higher binding affinity to Tau and prevents cleavage and aggregation of Tau *in vitro*

(A) Biacore analysis demonstrated the interaction ApoE3 WT or ApoE3 R136S with Tau proteins. A two-tailed t test was performed for K_d value, and data are shown as means ± SEM. $n = 3$ each group; $^*p < 0.05$.

(B) SY5Y cells were co-transfected with GST-Tau and GFP-tagged control vector, ApoE3, or ApoE3 R136S mutant. GST pull-down and immunoblot of lysates were conducted. Quantification of immunoblots showed that ApoE3 R136S displayed higher binding affinity to Tau than ApoE3.

(C and D) SY5Y cells were treated with DOPEGAL (60 μM) after co-transfection with GST-Tau (~75 kDa) and GFP-tagged control vector, ApoE3, or ApoE3 R136S mutant (~55–60 kDa). Immunoblot showed that Tau oligomerization and phosphorylation by DOPEGAL were decreased by ApoE3 R136S. GFP-ApoE3 significantly elevated AEP enzymatic activities versus GFP control or GFP-ApoE3 R136S.

(legend continued on next page)

occipital cortex, indicating that the Christchurch variant may impact the distribution of Tau pathology, modulate age at onset, severity, progression, and clinical presentation.[6] Nevertheless, the molecular mechanism of how ApoE3 R136S mutant regulates Tau pathology and its propagation in the brain remains unclear.

In the current work, we demonstrate that ApoE3 R136S mutant binds to Tau with stronger affinity than ApoE3 wild type (WT), decreasing its uptake into neurons and microglia. Moreover, the R136S mutant suppresses asparagine endopeptidase (AEP) activation, which is a critical protease acting as a delta-secretase that cleaves both APP and Tau and promotes Aβ and NFT pathologies in AD.[7–9] Consequently, R136S blunts Tau N368 fragmentation by active AEP and reduces Tau hyperphosphorylation and aggregation, abrogating its neurotoxicity. Remarkably, R136S prominently diminishes Tau pathology propagation in both 5xFAD and Tau P301S mouse models, restoring the cognitive functions.

RESULTS

ApoE3 R136S binds to Tau and protects it from active AEP cleavage, suppressing Tau aggregation in cells

The observations that the Christchurch variant R136S diminishes Tau pathology distribution in the frontal cortex and hippocampus (HC), slowing the disease progression and clinical presentation,[6] prompt us to hypothesize that this mutant may directly impact Tau biochemical activities via physical interaction. To explore this possibility, we conducted a Biacore binding assay and found that the R136S mutant revealed stronger binding affinity to Tau than ApoE3 WT with K_d approximately 0.1 μM versus 0.8 μM, respectively (Figure 1A). Co-immunoprecipitation (coIP) from glutathione S-transferase (GST)-Tau and GFP-ApoE3 co-transfected SH-SY5Y cells exhibited similar results (Figure 1B). To examine whether more robust interaction influences Tau proteolytic fragmentation by AEP, which cuts Tau at N368 and promotes its hyperphosphorylation and aggregation,[7] we treated the co-transfected cells with 3,4-dihydroxy-phenylglycolaldehyde (DOPEGAL), a reactive metabolite from norepinephrine (NE) by monoamine oxidase-A (MAO-A), and observed evident AEP activation under both control and ApoE3 WT, which was profoundly inhibited in the presence of ApoE3 R136S. Consequently, Tau N368 fragmentation and p-Tau signals revealed by AT8 and Tau aggregation were clearly reduced in cells co-transfected with R136S as compared with control or ApoE3 WT. Fitting with our previous reports[10,11] that Tau K353 residue covalent modification by DOPEGAL activity was soothed by R136S versus control or ApoE3 WT cells (Figures 1C and 1D). To further explore whether the tight physical interaction may impact Tau cleavage by active AEP, we performed an *in vitro* assay with purified recombinant Tau and ApoE3 proteins. Compared with vehicle control, ApoE3 WT decreased Tau fragmentation by AEP, which was further repressed by R136S, suggesting that the latter possesses a stronger protective effect than the former (Figure 1E). Cellular aggregation assays with Tau repeat domains in stably transfected HEK293 cells showed that Tau pre-formed fibrils (PFFs) induced pronounced Tau-aggregated inclusions in the presence of ApoE3 WT, which were strongly suppressed by R136S mutant (Figure 1F). Hence, ApoE3 R136S strongly binds to Tau and protects it from AEP cleavage, repressing Tau aggregation.

ApoE3 R136S blunts Aβ or Tau PFF-induced AEP activation, Tau aggregation, and neuronal cell death in primary cortical neurons

Next, we extended the *in vitro* biochemical studies into primary cortical neurons, which were infected with virus expressing ApoE3 WT or R136S, followed by adeno-associated virus (AAV)-Tau infection in the presence of pre-aggregated Aβ42. Consistent with the findings in co-transfected SH-SY5Y cells, coIP demonstrated R136S bound more Tau than ApoE3 WT. Immunoblotting showed that Aβ treatment elicited significantly more AEP activities in the presence of ApoE3 WT than R136S; accordingly, the levels of Tau N368 fragmentation, a pathological truncate in AD patient brain,[12,13] tightly correlated with upstream protease activities (Figures 2A and 2B). The secreted Aβ42 levels in the culture media also oscillated with AEP enzymatic activities, fitting with our previous report that AEP acts as delta-secretase and dictates Aβ production in neurons and in the brain[8] (Figure 2C). LDH assay revealed that R136S significantly decreased Tau/Aβ-induced neuronal cell death compared with control or ApoE3 WT (Figure 2D).

To further examine whether exogenously introduced lipidated ApoE3 R136S exerts its neural protection similarly to the findings above, we pretreated primary neurons with lipidated ApoE proteins, followed by either pre-aggregated Aβ42 or Tau PFFs or their combination. Lipidated ApoE3 WT and R136S proteins and human Tau recombinant proteins were purified and validated by Coomassie blue staining. Notably, ApoE3 WT and R136S recombinant proteins demonstrated opposite cholesterol uptake and efflux effects. GFP tag did not affect ApoE3 secretion, half-life, or lipidation (Figure S1). Immunoblotting showed that Aβ and Tau PFF mixture induced stronger AEP activities than Aβ or Tau PFFs alone under vehicle control, and these effects were further strengthened in the presence of lipidated ApoE3 WT. By contrast, AEP activation was substantially blunted by lipidated R136S (Figure 2E, 6th panel). Consequently, Tau N368 cleavage and Tau oligomerization revealed by anti-T22 were more prominent in neurons treated with ApoE3 WT than vehicle control, and these biochemical events were dramatically blocked by R136S (Figure 2E, 4th and 5th panels). Noticeably, total Tau (Tau 5) and p-Tau (AT8) signals were much stronger in neurons treated with Tau PFFs, and these

(E) Recombinant Tau (5 μg) and active AEP (0.05 μg) proteins were incubated in the presence of ApoE3 WT or ApoE3 R136S proteins. Immunoblot and quantification demonstrated that cleavage of Tau by AEP was inhibited by ApoE3 R136S.
(F) HEK293 cells stably transfected with GFP-Tau RD were exposed to control, lipidated ApoE3, or ApoE3 R136S proteins, and then followed administration of Tau PFFs. Representative images showed the Tau inclusions in cells, and quantification demonstrated the inhibition of Tau inclusions by ApoE3 R136S. Scale bar, 20 μm. One-way (B–E) or two-way ANOVA (F) was used for statistical analysis, and data are shown as mean ± SEM. $n = 3$ (B, C, and E), $n = 4$ (D), $n = 5$ (F) per group. $^*p < 0.05$, $^{**}p < 0.01$.

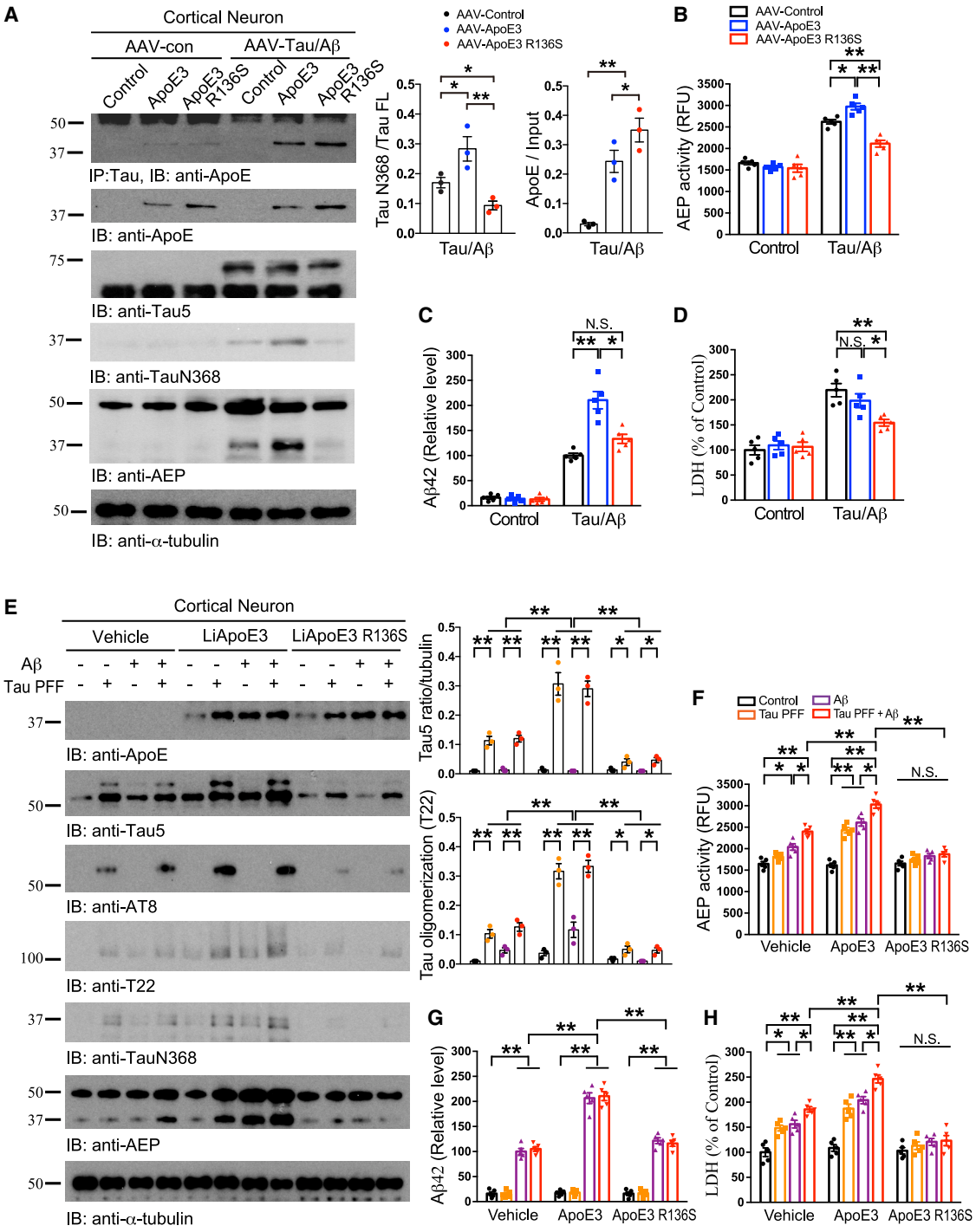


Figure 2. ApoE3 R136S displays the protective effect against Aβ or Tau PFF-induced toxicity in neurons

Primary neurons were cultured and infected with AAV-control, AAV-ApoE3, or ApoE3 R136S, followed by coinfection of AAV-Tau (2N4R) in the presence of aggregated Aβ.

(A) Western blot analysis showed that ApoE3 R136S inhibited Tau cleavage by AEP with the quantification of anti-Tau N368 (~37 kDa) and immunoprecipitated anti-ApoE (~34 kDa).

(B) The activation of AEP was confirmed by enzymatic assay.

(C) The change of Aβ levels by ApoE3 was measured by ELISA.

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effects were more potent by lipidated ApoE3 than by the vehicle control. By contrast, these activities were robustly suppressed by R136S (Figure 2E, 2nd and 3rd panels). Quantification of Tau inclusion and oligomerization is shown on the right. The AEP enzymatic assay from the neuronal lysates mirrored active AEP fragment in western blotting (Figure 2F). Aβ ELISA showed that lipidated ApoE3 and Aβ-cotreated neurons secreted significantly more abundant Aβ42 than vehicle control or R136S. Again, R136S evidently reduced neuronal cell death compared with control or ApoE3 WT (Figures 2G and 2H). As shown in Figure S2, both GFP-ApoE3 and ApoE3 demonstrated the same activity in mediating Tau PFF-triggered AEP signaling and neuronal cell death, indicating that GFP tag does not affect ApoE3 biological activity. Therefore, ApoE3 R136S strongly blocks AEP activation and Tau N368 fragmentation and aggregation, protecting neurons from cell death triggered by Aβ or Tau PFFs.

ApoE3 R136S reduces Tau PFF uptake into noradrenergic neurons and neural cell death

The pre-tangle form of Tau appears in the locus coeruleus (LC), the major central source of NE, before any other area of the brain, making it the earliest detectable AD-like neuropathology.[14–16] To assess whether R136S affects Tau PFF uptake and neurotoxicity, we prepared the noradrenergic neurons and treated them with pre-aggregated fluorescent-tagged Tau or Aβ fibrils in the presence of ApoE3 WT or R136S proteins. Remarkably, lipidated ApoE3 R136S strongly blocked both Aβ and Tau PFF uptakes (Figures 3A–3C). Co-staining of oligomeric Tau antibody, T22, or thioflavin S (ThS) with Tau5 demonstrated that the fluorescent-labeled Tau in Figure 3A represented the aggregated form of Tau, as compared with untagged endogenous Tau (Figure 3D). Terminal deoxynucleotidyl transferase-mediated dUTP nick end-labeling (TUNEL) staining revealed that R136S substantially protected LC neurons from apoptosis triggered by the mixture of Aβ and Tau PFFs (Figures 3E and 3F).

Lipoprotein receptor-related protein 1 (LRP1), one of the family members of LDL-like receptors, acts as a master regulator for Tau uptake and propagation.[17] *In vitro* studies showed that lipid-bound ApoE3 R136S displays a reduction of 60% in LDL receptor binding capacity as compared with WT ApoE3.[18] To examine whether this receptor plays any role in mediating Tau PFFs neuronal uptake in the presence of ApoE3, we treated the LC neurons with ATTO-labeled Tau PFFs and lipidated ApoE3 proteins in the presence of receptor-associated protein (RAP), a potent inhibitor of ligand binding to LRP1 extensively used to inhibit the functional activity of these LDL receptor family members. RAP significantly reduced Tau internalization into neurons treated with ApoE3 compared with control, and the inhibitory effect was further enhanced by R136S (Figures 3H and 3I). Immunoblotting showed that RAP potently blocked AEP activation in ApoE3 neurons, which was further antagonized by R136S. Consequently, p-Tau and K353-DOPEGAL-modified Tau activities coupled with upstream AEP activities, whereas both ApoE and LRP1 levels were equal among the groups (Figure 3G). Immunoblotting showed that the activation of LRP1 downstream kinases, including dual leucine zipper kinase (DLK) and extracellular response kinase (ERK), by lipidated ApoE3 was weakened by lipidated ApoE R136S both were inhibited by RAP (Figures S2E–S2H). Thus, R136S strongly blocks Tau uptake into LC neurons and protects against its neurotoxicity.

ApoE3 R136S attenuates Tau uptake and inhibits C/EBPβ/AEP signaling activation in microglial cells

Microglia, the innate immune cells of the brain, are major players in AD pathogenesis. Microglia are long-lived, tissue-resident professional phagocytes that survey and rapidly respond to changes in their microenvironment.[19] Aβ or Tau oligomers or fibrils propagate to adjacent microglia or neurons through exosomes and change the neuroinflammatory microenvironment by microglial metabolic reprogramming.[20] To investigate whether ApoE R136S mediates microglial endocytosis of Aβ and Tau and their metabolism within the microglial cells, we pretreated the HMC3 microglial cell line with fluorescent-labeled Aβ or Tau aggregates for 2 h. Fluorescent Aβ uptake was significantly elevated in the presence of R136S compared with ApoE3 WT. Aβ ELISA analysis of cell lysates correlated with this format (Figures 4A–4C). On the other hand, compared with ApoE3 WT, R136S substantially decreased Tau uptake (Figures 4E–4G). To further explore the effect of R136S on microglial metabolism of the endocytosed aggregates, we treated HMC3 cells with the mixture of Aβ and ApoE3 proteins in the medium for 2 h. After washout, the microglia were cultured in the regular medium for another 18 h, followed by immunofluorescent (IF) co-staining for the aggregated Aβ and ApoE complexes remained in microglia cells. Once again, R136S strongly escalated Aβ internalization into microglia and its intracellular concentrations compared with ApoE3 WT (Figure 4D). We made the similar observation with Tau to its uptake (Figure 4H).

Previous study shows that ApoE regulates cholesterol uptake and efflux.[21] Accordingly, we demonstrated that lipidated ApoE3 R136S recombinant proteins blocked cholesterol uptake into HCM3 cells compared with ApoE3 WT. By contrast, they significantly elevated cholesterol efflux from the cells (Figures S1C and S1D). No matter whether microglia cells were treated with 2 h or 2 h/18 h washout paradigm, R136S prominently reduced proinflammatory cytokines, including interleukin (IL)-1β, IL-6, and tumor necrosis factor alpha (TNF-α), released from microglial cells as compared with ApoE3 WT (Figures S3A–S3F).

(D) LDH assay showed that ApoE3 R136S reduced cell death by Tau and Aβ. One-way ANOVA was used for statistical analysis, and data are shown as mean ± SEM. $n = 3$ for (A) and $n = 5$ for (B)–(D). $^{*}p < 0.05$, $^{**}p < 0.01$. Primary neurons were cultured and treated with vehicle, lipidated ApoE3, or lipidated ApoE3 R136S, followed by addition of Tau PFF and Aβ.

(E) Western blot analysis showed that lipidated ApoE3 R136S inhibited Tau expression and oligomerization.

(F) The activation of AEP was confirmed by enzymatic assay.

(G) The change of Aβ levels by ApoE3 or ApoE3 R136S was measured by ELISA.

(H) LDH assay showed that ApoE3 R136S reduced cell death by Tau and Aβ. Two-way ANOVA was used for statistical analysis, and data are shown as mean ± SEM. $n = 3$ for (E), $n = 5$ for (F)–(H). $^{*}p < 0.05$, $^{**}p < 0.01$.

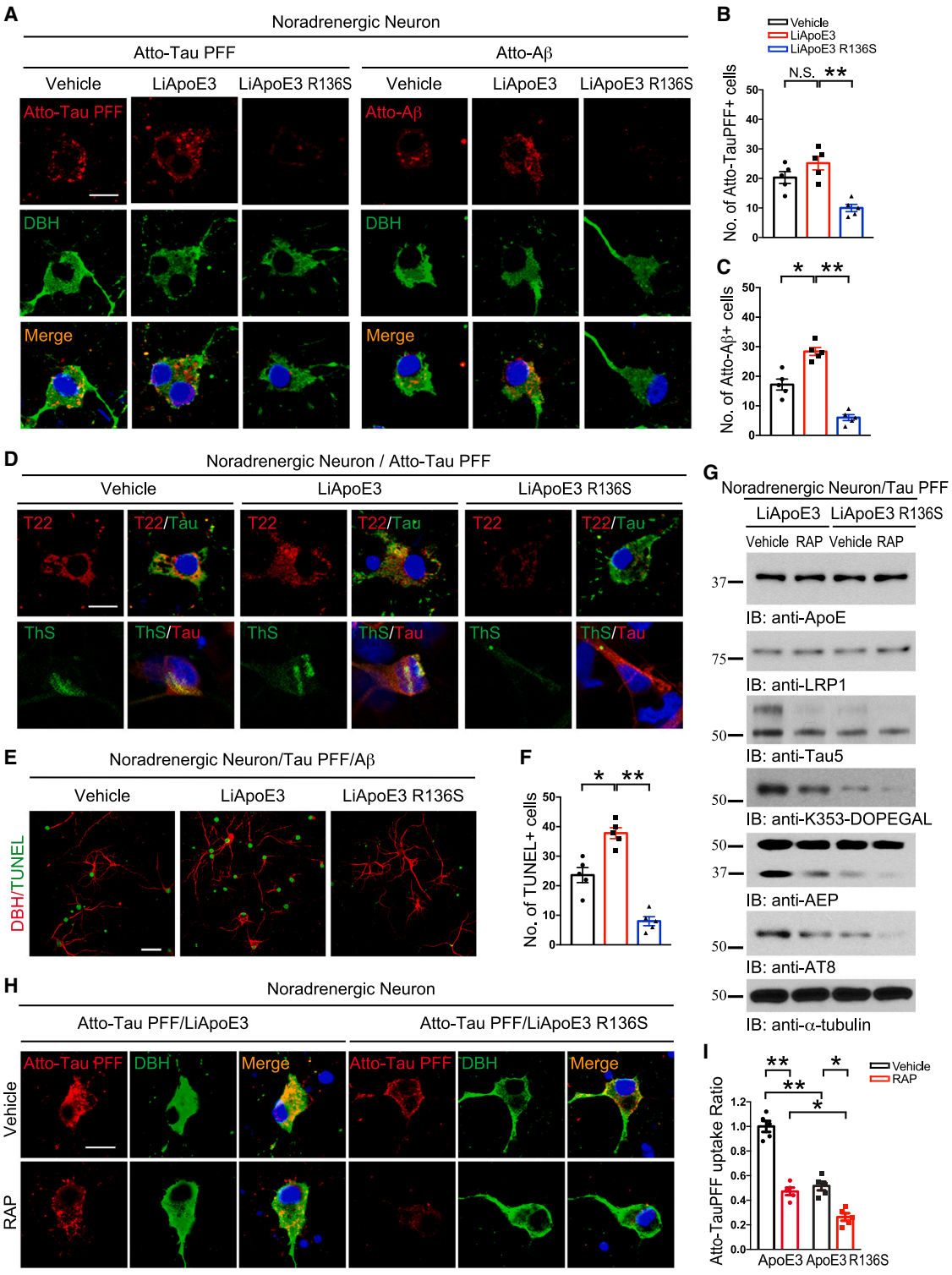


Figure 3. ApoE3 R136S inhibits uptake of Aβ or Tau PFFs into NE neurons and neuronal death

Primary LC neurons were prepared and treated with Atto-Tau PFF (2N4R) or Atto-Aβ in the presence of vehicle, lipidated ApoE3, or lipidated ApoE3 R136S.

(A) Upper panels showed representative immunofluorescent images for uptake of Atto-labeled Tau PFF or Aβ into neurons (red). Second panels showed NE neurons for DBH (green). Scale bar, 20 μm.

(B and C) Quantification of Tau PFF (B) or Aβ+ (C) cells demonstrated that neuronal internalization of Tau PFF or Aβ was inhibited by ApoE3 R136S. One-way ANOVA was used for statistical analysis, and data are shown as mean ± SEM. $n = 5$. *$p < 0.05$, **$p < 0.01$.

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ApoE3 R136S attenuates proinflammatory cytokines and neurotoxic exosomal vesicles secreted by microglia

Activated microglia migrate toward a lesion and phagocytose of cellular debris or damaged cells, orchestrating neuroinflammation by secreting important inflammatory mediators. To interrogate the effect of ApoE3 R136S in-depth on these mediators in the presence of aggregated Aβ or Tau, we conducted the blot assay loaded with a variety of cytokines and chemokines with the media secreted from HCM3 cells treated with ApoE3 WT or R136S, respectively. Under Aβ treatment, CXC-chemokine ligand (CXCL)1, IL-6, and IL-8 levels were significantly repressed by R136S as compared with ApoE3 WT. By contrast, substantially more effectors, including C-C motif ligand (CCL)2, CXCL12, INF-γ, and IL-1β and TNF-α levels, were drastically suppressed by R136S, in addition to the above 3 mediators (Figures 4I–4L).

AEP is robustly activated and shreds Tau into fragments after endocytotic uptake into active microglia.[22] Exosomes can transport Aβ, Tau, and cytokines between the cells and alter the physiology of recipient cells.[22,23] They can also assist in Aβ clearance and regulation of synaptic activity.[24] Fluorescent staining of the secreted extracellular vesicles showed that R136S notably diminished secreted exosomes that contained the released AEP as compared with ApoE3 WT. The exosomes were characterized by their biomarker Tsg101 (Figures 5A and 5B). Immunoblotting of these extracellular vesicles further validated the reduced Aβ or Tau oligomers or aggregates and AEP in the decreased exosomes secreted from microglia treated with R136S as compared with ApoE3 WT (Figures 5C and 5D).

We have reported that CCAAT/enhancer binding protein beta (C/EBPβ), an Aβ- or inflammatory cytokine-activated transcription factor, dictates AEP transcription and mediates AD pathogenesis.[24,25] C/EBPβ augmentation in microglia increases neuroinflammation and accelerates Tau propagation and neurodegeneration in AD mouse model.[26] Western blotting analysis showed that the C/EBPβ upstream effectors of p-RAC-alpha serine/threonine-protein kinase (Akt) and p-Erk1/2 signals were both reduced by R136S versus ApoE3 WT no matter whether the cells were cotreated with Aβ or Tau. Consequently, p-C/EBPβ and C/EBPβ levels were obviously decreased, as was the downstream target AEP (Figures S3G and S3H). Activation of neutral sphingomyelinase (nSMase) is an established part of the response of cytokines such as TNF-α. Exosomes stimulate aggregation of Aβ42 *in vitro* and *in vivo* and interfere with uptake of Aβ. Exosome secretion is regulated by nSMase2, a key regulatory enzyme generating ceramide from sphingomyelin.[27,28] Quantification of nSMase2 enzymatic activities revealed that R136S steadily suppressed Aβ or Tau aggregate-induced nSMase2 activities in microglia cells compared with ApoE3 WT (Figures S3I and S3J). Treatment with primary neurons with the secreted exosomes from ApoE3 WT in the presence of Aβ or Tau elicited evident caspase-3 activation or TUNEL signals associated with extensive death, which were manifestly blocked by R136S (Figures 5E and 5F). Immunoblotting confirmed that AEP substrates Tau N368 and UNC5C N467 fragmentation and caspase-3 activation were sharply decreased (Figures 5G and 5H). T835M mutation in netrin receptor UNC5C predisposes to the late-onset AD and increases neuronal cell death.[29] We have previously reported that truncates UNC5C and elevates its neurotoxicity, contributing to AD pathogenesis.[30] Consistently, AEP enzymatic assay also supported that AEP was markedly affected by R136S versus ApoE3 WT (Figures 5I and 5J). Collectively, these data indicate that ApoE3 R136S greatly inhibits C/EBPβ/AEP signaling and suppresses proinflammatory cytokines from microglia triggered by Aβ or Tau aggregates. Correspondingly, exosomal neurotoxicity is also reduced by R136S.

ApoE3 R136S reduces C/EBPβ/AEP signaling activation and prevents neural loss in the HC of 5xFAD and Tau P301S mice

Mounting evidence suggests that C/EBPβ/AEP signaling drives AD pathogenesis.[31–33] The PSEN1 E280A mutation patient carries two copies of ApoE3 R136S mutant and displays severely elevated brain Aβ with limited Tau pathology and neurodegeneration.[5] To explore whether R136S blocks C/EBPβ/AEP signaling, impacting Aβ pathology, we injected AAV-ApoE3 WT or R136S virus into the HC of 5xFAD mice at 3 months old. After 3 months, immunohistochemistry (IHC) with anti-Aβ revealed that Aβ plaques were prominently decreased by R136S as compared with ApoE3 WT (Figures S4A and S4B). IF staining corroborated IHC staining results. Interestingly, ionized calcium-binding adapter molecule 1 (Iba-1) activities in R136S-injected brains were greatly reduced, whereas neuronal nuclei (NeuN) signals were inversely escalated (Figures S4C–S4G), fitting with the findings that microglia Aβ uptake increase and exosome neurotoxicity reduction by R136S (Figures 4 and 5). Accordingly, both AEP and nSMase enzymatic activities were significantly reduced (Figures S4H and S4I). We conducted immunoblotting and found that R136S strongly inhibited C/EBPβ/AEP signaling, blocking cleavage of the AEP substrate UNC5C N467. Noticeably, active caspase-3 was also robustly diminished (Figure S4J). Quantification with Aβ40 and Aβ42 ELISA also validated these observations (Figures S5A–S5D). In alignment with reduced microglial activation, inflammatory cytokines, including IL-1β, IL-6, and TNF-α levels, were significantly decreased by R136S compared with ApoE3 WT (Figures S5E–S5G). Comparable ApoE3 WT and

(D) Representative immunofluorescent images for oligomeric Tau (T22, red) and endogenous Tau, including untagged Tau (green), in the upper panel. The lower panel shows aggregated Tau, thioflavin S (ThS, green), and Tau (red).

(E and F) Immunofluorescent and quantification of TUNEL+ cells demonstrated that cell death by Tau PFF and Aβ was inhibited by ApoE3 or R136S. One-way ANOVA was used for statistical analysis, and data are shown as mean ± SEM. $n = 5$. $^{*}p < 0.05$, $^{**}p < 0.01$. Lipoprotein receptor-related protein 1 (LRP1) inhibitor RAP blocks neuronal internalization of Tau fibrils. Primary LC neurons were prepared and treated with Atto-Tau PFFs and lipidated ApoE3 in the presence of vehicle or RAP (500 nM) recombinant proteins, an antagonist of lipoprotein receptor-related protein 1 (LRP1).

(G) Western blot analysis showed that RAP blocked the uptake of Tau into neurons.

(H) Representative immunofluorescent images for Atto-Tau PFF (red) and DBH (green). Scale bar, 20 μm.

(I) Quantification of uptake of Tau PFFs demonstrated that neuronal internalization of Tau PFF was prohibited by RAP. Two-way ANOVA was used for statistical analysis, and data are shown as mean ± SEM. $n = 5$. $^{*}p < 0.05$, $^{**}p < 0.01$.

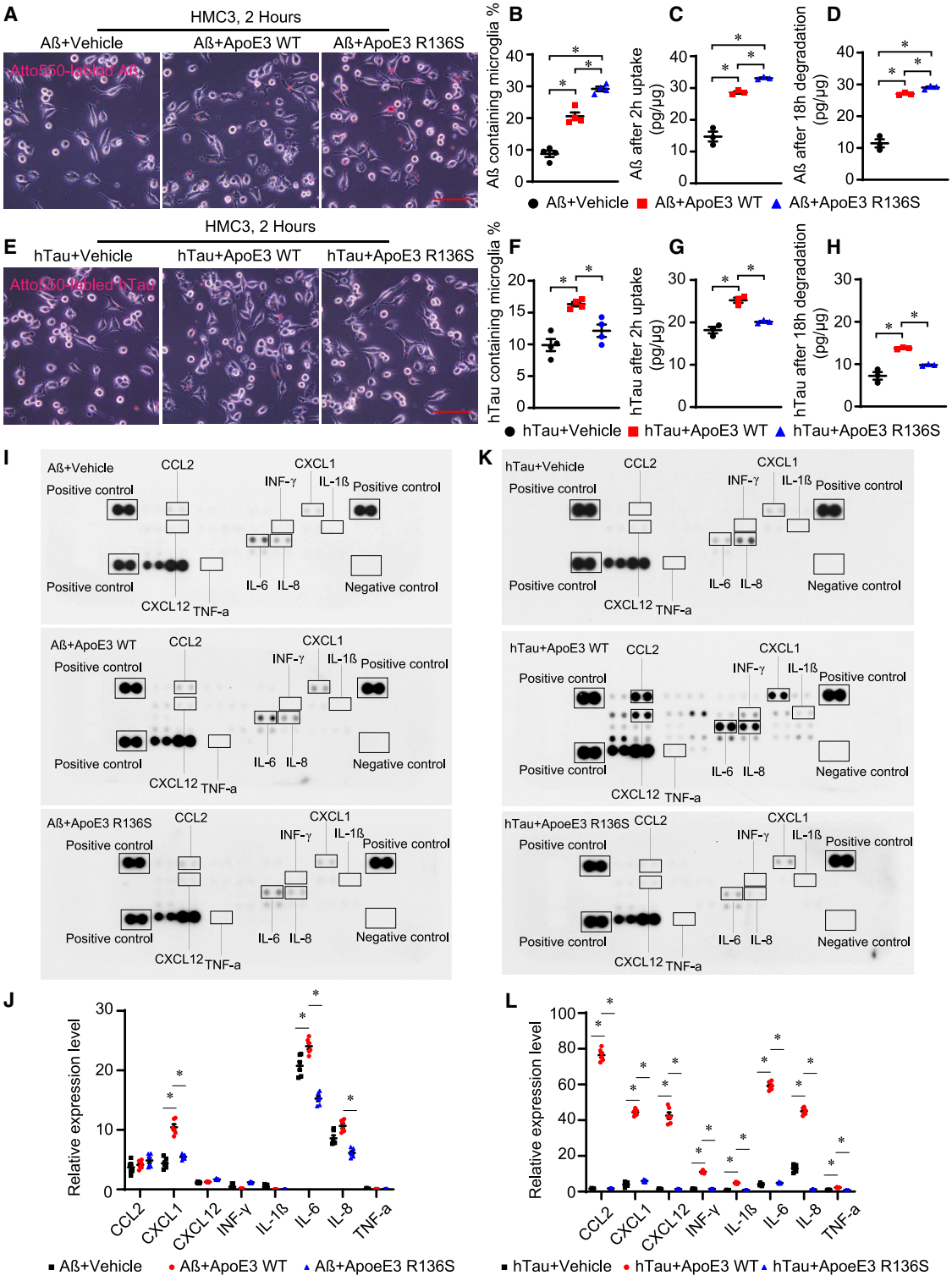


Figure 4. ApoE3 R136S promotes uptake of Aβ and decreases secretion of inflammatory factors by microglia

(A) Uptake of the Aβ-ApoE3 complex by HMC3 cells after incubation for 2 h. Atto550-labeled Aβ aggregates were incubated with WT and R136S mutant ApoE3 and then added to the medium of HMC3 cells. The images were captured after incubation for 2 h. Scale bar, 200 μm.

(B) Quantification of HMC3 cells containing atto550-labeled Aβ.

(C) ELISA analysis of Aβ in HMC3 cells after incubation for 2 h.

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R136S levels were found by immunoblotting (IB) and quantitative reverse-transcription PCR (RT-qPCR) analysis (Figure 5H; Figure S4J). Morris water maze (MWM) behavioral tests demonstrated that R136S significantly increased 5xFAD mice learning and memory functions in comparison with ApoE3 WT, while it did not affect the motor function since the swim speed was not altered by R136S (Figures S4K–S4M). Fear conditioning tests also showed that R136S greatly elevates the cognitive functions in opposition to ApoE3 WT (Figures S4N and S4O). Therefore, R136S strongly alleviates neuroinflammation and protects neurons from degeneration in 5xFAD mice, ameliorating cognitive impairments.

To investigate the effect of R136S mutant on Tau pathology, we injected AAV-ApoE3 WT or R136S virus into the HC of Tau P301S mice at the age of 3 months. After 3 months, we performed IHC staining with AT100 antibody and p-Tau 181 ELISA quantification, which confirmed that R136S profoundly abated p-Tau activities in comparison with ApoE3 WT (Figures 6A–6C). IF staining showed that AT8 and Iba-1 signals were steadily diminished; consequently, NeuN levels were inversely augmented (Figures 6D–6H). Both AEP and nSMase2 enzymatic activities were decreased by R136S in contrast to ApoE3 WT (Figures 6I and 6J). Immunoblotting showed that R136S vigorously blocked C/EBPβ/AEP signaling and blunted Tau N368 fragmentation. Again, caspase-3 activation was also sturdily suppressed (Figure 6K). As expected, inflammatory cytokines were also dwindled by R136S as compared with ApoE3 WT (Figures S6A–S6C). T22 staining showed that ApoE3 R136S significantly reduced Tau aggregation in Tau P301S mice (Figures S6D and S6E). Accordingly, both MWM and fear conditioning tests supported that R136S primarily increased the cognitive functions in contrast to ApoE3 WT (Figures 6L–6P). Hence, ApoE3 R136S antagonizes C/EBPβ/AEP pathway, attenuates Tau pathology, and prevents neural loss in Tau P301S, alleviating cognitive dysfunctions. To exclude the effect of mouse ApoE gene, we used Tau P301S/ApoE-knockout mouse model to show the effect of ApoE3 R136S on Tau pathology (Figure S7). We injected AAV-control, AAV-ApoE3 WT, or AAV-R136S virus into the HC of Tau P301S/ApoE−/− mice at the age of 6 months. After 1 month, we performed immunoblotting and found that R136S vigorously blocked C/EBPβ/AEP signaling and blunted Tau N368 fragmentation (Figures S7A and S7B). IF staining showed that AT8 and Iba-1 signals were steadily diminished; consequently, NeuN levels were conversely elevated (Figures S7C–S7F).

ApoE3 R136S blocks Tau pathology spreading from the LC to the forebrain in 5xFAD mice

Most recently, we have reported that NE metabolite DOPEGAL activates AEP and pathological Tau aggregation in the LC.[10,34] AEP is indispensable for Tau propagation from the LC to the forebrain regions.[11] To explore the potential effect of ApoE3 R136S in this process, we co-injected Tau PFFs with AAV-ApoE WT or R136S virus in the LC of 5xFAD mice at the age of 3 months old, respectively. After 3 months, we conducted IF co-staining with anti-AT8 and anti-dopamine β-hydroxylase (DBH), which specifically stained the crucial enzyme DBH in noradrenergic neurons. Noticeably, DBH-positive LC neurons were significantly lost, whereas AT8 activities were inversely elevated in the brain injected with ApoE3 WT as opposed to control or R136S. By contrast, these signals remained unchanged under vehicle control (Figures 7A–7D). Moreover, ApoE and Tau were shown to colocalize in LC neurons (Figure 7B). Although Aβ40 concentrations were comparable among the groups, Aβ42 levels were significantly higher in ApoE3 WT groups than in the control or R136S groups (Figures 7E and 7F). AT8-positive Tau pathologies were pronouncedly spread from the LC to EC to HC in ApoE3 WT-injected brains in contrast to control or R136S (Figure 7G). Consequently, MWM and fear conditioning tests demonstrated that learning and memory activities were substantially impaired in 5xFAD mice injected with ApoE3 WT in comparison with control or R136S (Figures 7H–7L). Therefore, ApoE R136S decreases Tau pathology spreading from the LC to the forebrain, ameliorating cognitive disorders in 5xFAD mice compared with ApoE3 WT.

DISCUSSION

Hyperphosphorylated Tau appears exclusively in the LC in Braak stage 0 stages, spreads to the entorhinal cortex (EC) in Braak stages I and II, then propagates to the HC and frontal cortex in stages III–VI, and the LC is one of the origins of Tau pathology in AD.[12,35] Injection of Tau pre-formed filaments (PFFs) or Tau virus into the LC leads to progressive loss of LC neurons and the appearance of Tau pathology in the forebrain.[10,34,36,37] Based

(D) ELISA analysis of Aβ in HMC3 cells after degradation for 18 h. The HMC3 cells after the 2-h phagocytosis were washed with PBS 3 times and followed by an 18-h incubation in conditioned medium without serum. Then the cells were harvested at 20 h for ELISA test.
(E) Uptake of the Tau-ApoE3 complex by HMC3 cells after incubation for 2 h. Atto550-labeled hTau aggregates were incubated with WT and R136S mutant ApoE3 and then added to the medium of HMC3 cells. The images were captured after incubation for 2 h. Scale bar, 200 μm.
(F) Quantification of HMC3 cells containing atto550-labeled hTau.
(G) ELISA analysis of hTau in HMC3 cells after incubation for 2 h.
(H) ELISA analysis of hTau in HMC3 cells after degradation for 18 h. The HMC3 cells after the 2-h phagocytosis were washed with PBS 3 times and followed by an 18-h incubation in conditioned medium without serum. Then the cells were harvested at 20 h for ELISA test.
(I) Representative dot blots of neuroinflammatory factors in the conditioned medium after 18-h degradation for treated with Aβ-ApoE3 or Aβ-ApoE3 R136S complex. Cytokine levels were detected using the Proteome Profiler Human Cytokine Array.
(J) Quantification of expression levels of CCL2, CXCL2, CXCL12, IFN-γ, IL-β, IL-6, IL-8, and TNF-α after HMC3 cells were incubated with Aβ-ApoE3 or Aβ-ApoE3 R136S complex.
(K) Representative dot blots of neuroinflammatory factors in the conditioned medium after 18-h degradation for treated with hTau-ApoE3 or hTau-ApoE3 R136S complex. Cytokine levels were detected using the Proteome Profiler Human Cytokine Array.
(L) Quantification of expression levels of CCL2, CXCL2, CXCL12, IFN-γ, IL-β, IL-6, IL-8, and TNF-α after HMC3 cells were incubated with hTau-ApoE3 or hTau-ApoE3 R136S complex. Two-way ANOVA was performed for the analysis, and data are shown as mean ± SEM. $n = 4$. $^{*}p < 0.05$.

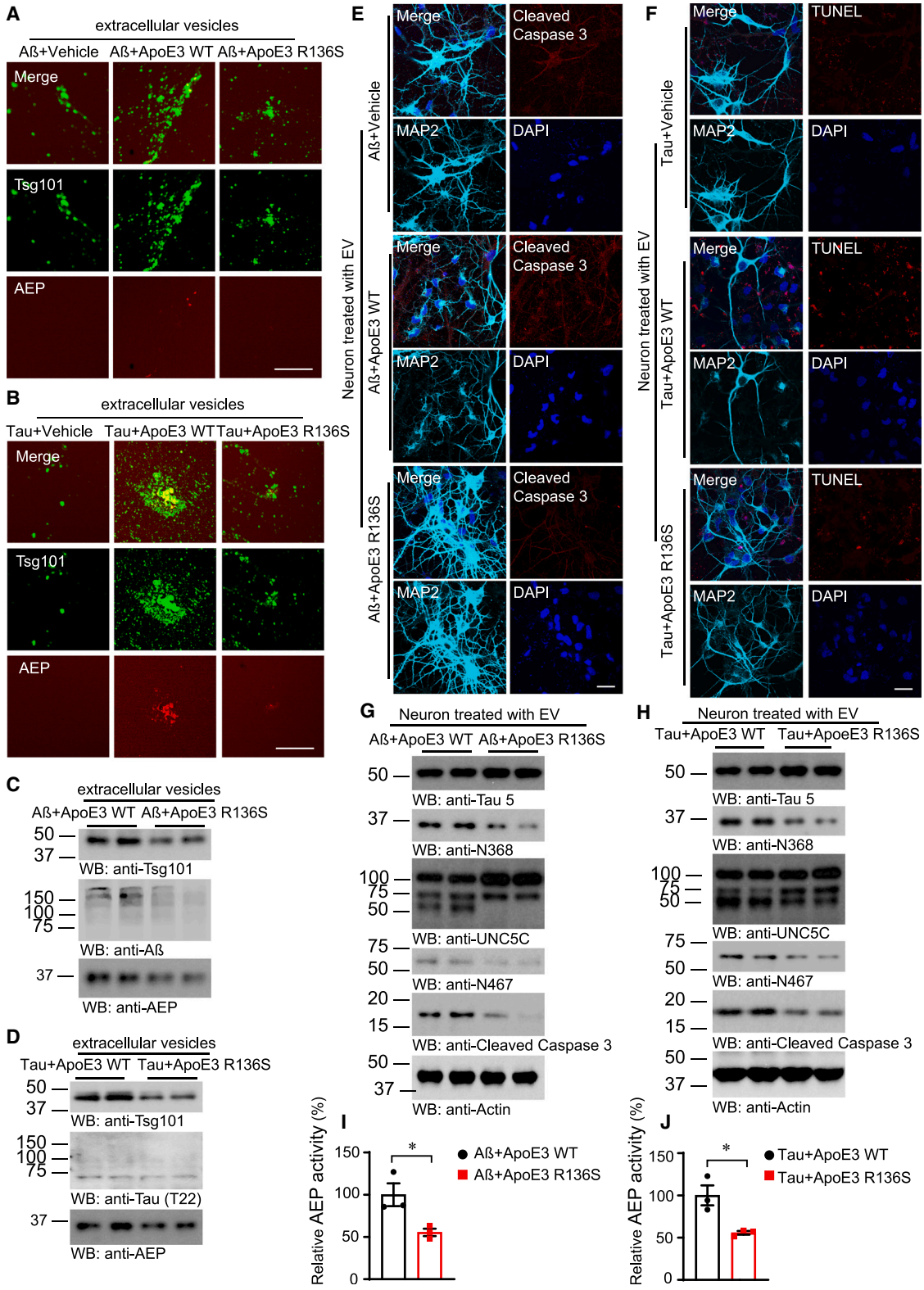


Figure 5. ApoE3 R136S inhibits Aβ/Tau secretion by microglia and ameliorates neuronal toxicity induced by Aβ and Tau

The conditioned medium without serum after the 18-h incubation from the uptake and degradation experiments (Figure 4) was collected and extracted for extracellular vesicles.

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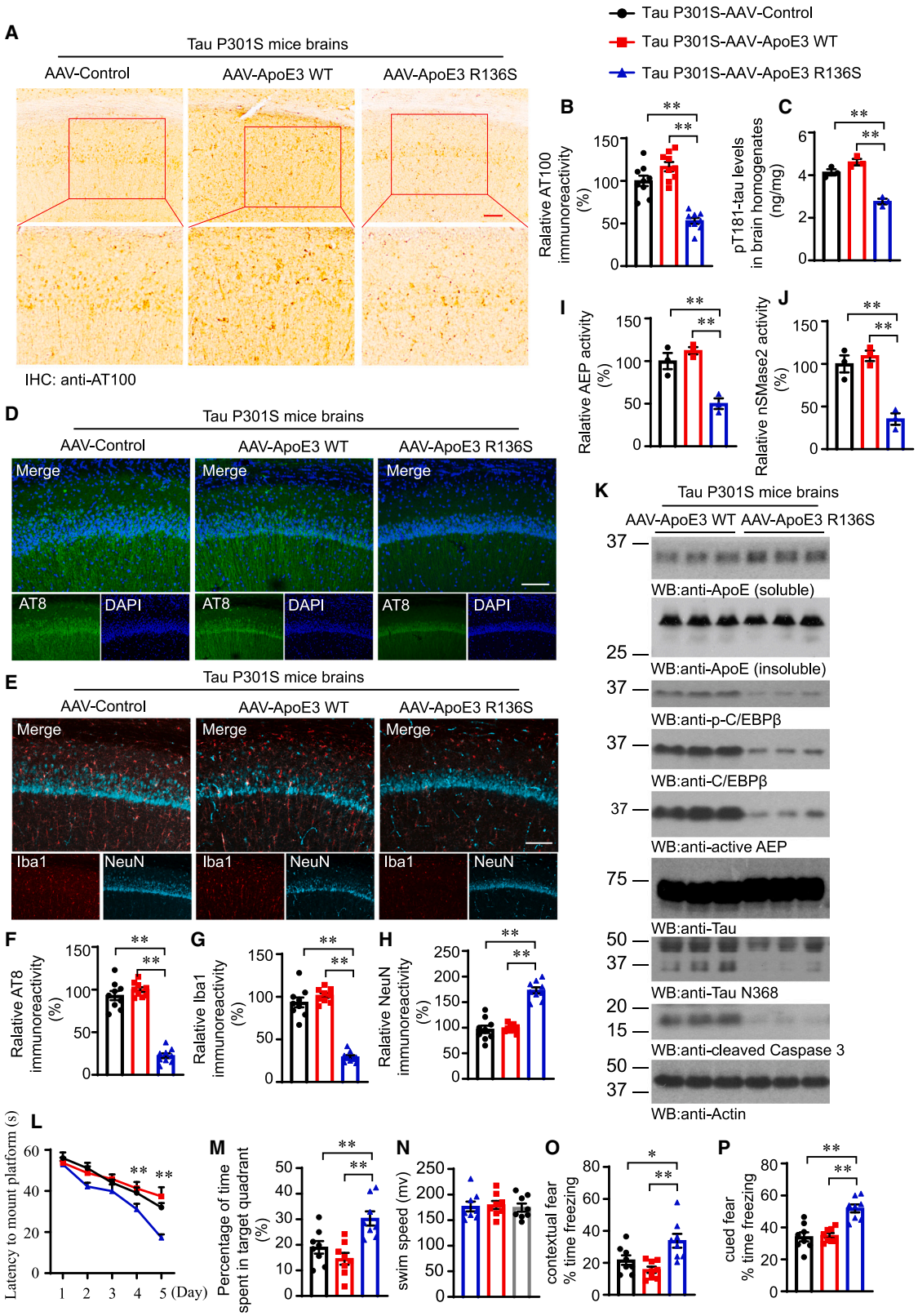
on the brain network foundation of Tau spreading and its association with using high-resolution transcriptomic genetic data, ApoE and glutamatergic synaptic genes, such as SLC1A2, have been suggested to play a central role in the spatial spreading of Tau.[38] The pathological accumulation of amyloid and Tau is not randomly distributed but rather follow a stereotypical spatial pattern that follows large-scale networks, suggesting that AD is a network-afflicting brain disease.[39,40] Pathological accumulation of Tau has been directly linked to longitudinal atrophy,[41] and is predictive of memory decline and clinical progression.[40,42] PSEN1 E280A carrier homozygous for the APOE3 Christchurch variant (R136S) was protected against AD symptoms for almost three decades. The patient displays a distinct anatomical pattern of Tau pathology, including atypical accumulation of *in vivo* Tau pathology. It has been suggested that APOE3ch may have a regionally specific role in modifying the effect of Tau on the severity, progression, and clinical presentation of AD, and that its protective role is associated with its expression and resulting impact on astrocytic and microglial function.[6] Previous studies support Tau pathology as the main driver of cognitive decline in PSEN E280A carriers, as suggested by other findings in this population,[43,44] and that the protection against this pathology should be clinically effective. The strong putative protective effect of homozygous APOE3ch in this patient suggests that ApoE may be one of the main modulators of Tau pathology in AD, possibly via different mechanisms, including aggregation by modifying ApoE heparin binding[5] and homeostatic regulation of glial cells.[6] In the past 2 years, several articles have revealed the protective role and mechanism of the APOE3 R136S mutation against AD from different angles.[45–47] It has been reported that homozygous R136S mutation rescued APOE4-driven Tau pathology, neurodegeneration, and neuroinflammation.[45] APOE3 R136S increased myeloid cell Tau uptake and degradation through APOE receptors.[46] Another research study suggested that ApoE3 R136S enhances resistance to Tau-related disease processes by downregulating the cyclic GMP-AMP synthase (cGAS)-stimulator of interferon genes (STING)-interferon (IFN) signaling pathway.[47]

In the current work, we show that the R136S mutant strongly binds to Tau and blocks Tau uptake via its receptor LRP1, attenuating its neurotoxicity (Figures 1, 2, 3, and 5). Expression of R136S in the HC of Tau P301S mice greatly diminishes microglial activation, neuroinflammation, Tau pathology, and neurodegeneration, resulting in alleviation of cognitive impairment (Figure 6). Expression of R136S in the HC of 5xFAD greatly reduced Aβ deposits and improved behavior. After injected with Tau PFF, ApoE R136S decreases Tau pathology spreading, ameliorating cognitive disorders in 5xFAD mice compared with ApoE3 WT (Figure 7; Figures S4–S6). Co-injection of R136S virus and Tau PFFs into the LC of 5xFAD mice profoundly decrease Tau pathology propagation, improving cognitive functions as opposed to ApoE3 WT (Figure 7). In this article, T22 was used to characterize oligomeric Tau, and THS was used to characterize aggregated Tau. The absence of silver staining or antibodies such as MC1 to characterize NFTs or pre-tangles is a limitation of the article.

It is true that some studies found that Tau PFFs cannot efficiently seed endogenous Tau in WT mice. In this study, however, we used a 5XFAD mouse model, which has extensive Aβ deposition in the brain. It is well established that Aβ promotes Tau aggregation. We clearly showed that exogenous Tau PFFs triggered Tau seeding, especially in mice overexpressing ApoE3 (Figure 7). This is consistent with our previous report that Tau PFFs induce the aggregation of endogenous mouse Tau when ApoE is overexpressed.[48] In that study, we showed that both human and mouse Tau PFFs can spread in our Thy1-ApoE4/C/EBPβ Tg mice, which do not contain any human Tau transgene. Mouse Tau aggregates in this mouse model are validated by Tau PET AV1451, and mouse Tau aggregates from this sporadic mouse model elicit Tau aggregation and Tau hyperphosphorylation (Figures 5 and 6). Moreover, our results fit with the most recent *Cell* paper's findings: R136S reduces hTau spreading in APP/PS1 mice, which also contain no hTau transgene.[46] In our study, 5xFAD mice were harvested for the samples at 6 months of age, 3 months after the AAV injection. Expression of ApoE3 R136S decreases Aβ burden in the 5xFAD mouse model, which is consistent with the recent report that showed that APP/PS1-21:APOE3 R136S females formed ∼20% fewer amyloid plaques compared with an APP/PS1-21:APOE3 female mouse cohort at 6 months of age.[46]

ApoE3 binds to Tau on the microtubule-binding domains, and S262 phosphorylation in repeat domain 1 decreases Tau binding to both microtubule and ApoE3.[49] Immunofluorescence image in Figure 7 demonstrates the colocalization of ApoE and Tau. However, the article has some limitations, including the absence of direct evidence from Biacore/equivalent analysis. Interestingly, the interaction between Tau and LRP1 is mediated by lysine residues in the microtubule-binding repeat region.[17] Recently, we reported that ApoE3 exhibits stronger protective effect toward Tau from cleavage by active AEP than ApoE4, which triggers AEP activation and subsequent Tau cleavage and phosphorylation by Tau N368 in primary neurons.[34] In this study, we demonstrate that ApoE3 R136S reveals

(A) The secretion of AEP via extracellular vesicles after HMC3 cells were incubated with Aβ-ApoE3 or Aβ-ApoE3 R136S complex. Scale bar, 30 μm.
(B) The secretion of AEP via extracellular vesicles after HMC3 cells were incubated with Tau-ApoE3 or Tau-ApoE3 R136S complex. Scale bar, 30 μm.
(C) Western blot showing the levels of Aβ and AEP in extracellular vesicles from HMC3 cells treated with Aβ-ApoE3 or Aβ-ApoE3 R136S complex.
(D) Western blot showing the levels of pathologic Tau and AEP secreted in extracellular vesicles from HMC3 cells treated with Tau-ApoE3 or Tau-ApoE3 R136S complex.
(E) Primary neurons were exposed to extracellular vesicles derived from HMC3 cells pretreated with Aβ-ApoE3 or Aβ-ApoE3 R136S complex. The morphology of neurons and the levels of active caspase-3 are shown. Scale bar, 20 μm.
(F) Primary neurons were exposed to exosomes derived from HMC3 cells pretreated with hTau-ApoE3 or hTau-ApoE3 R136S complex. The morphology of neurons and the levels of active caspase-3 were shown. Scale bar, 20 μm.
(G and H) Western blots showed the levels of AEP-cleaved Tau and UNC5C, as well as active caspase-3.
(I and J) AEP activities were quantified in primary neurons treated by Aβ-ApoE3 or Aβ-ApoE3 R136S complex (I) by hTau-ApoE3 or hTau-ApoE3 R136S complex (J). A two-tailed t test was performed, and data are shown as mean ± SEM. $n = 3$. $^*p < 0.05$.



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much stronger binding effect to Tau than ApoE3 WT and protects it from cleavage by active AEP in neuronal cells or primary neurons, triggered by DOPAGEL, a highly reactive and oxidative metabolite of NE, Aβ, or Tau PFFs (Figures 1 and 2). Tau uptake is a complex process that is not solely dependent on ApoE3. It is influenced by various factors, including different ApoE isoforms and several LDL receptors. Tau uptake can occur without ApoE3, and ApoE4 actually enhances Tau uptake more than other ApoE isoforms. In this context, ApoE3 R136S acts as a suppressor of Tau uptake (Figure 1). Accordingly, R136S potently suppresses Tau hyperphosphorylation and neurotoxicity (Figures 3 and 5).

In order to mimic the abundant Aβ in the brain of PTEN1 E280A carrier, we employed pre-aggregated Aβ *in vitro* neuronal model and 5xFAD, a well-established amyloid AD mouse model that exhibits extensive amyloid plaques across the forebrains. We found that R136S mutant displayed much stronger inhibitory effects on Tau neuronal uptake and Tau pathology propagation from the LC to EC to HC in 5xFAD mice than ApoE3 WT. Consequently, the cognitive impairment is significantly alleviated by R136S compared with ApoE3 WT (Figure 7). These *in vivo* observations are in alignment with the reported clinical findings. A carrier of the autosomal dominant AD PSEN1 E280A variant who is homozygous for the APOE3 Christchurch variant did not develop cognitive impairment until her seventies, 30 years after the typical age of clinical onset for her kindred.[5] Notably, it has been reported that a dozen individuals in Columbian family with PSEN1 mutation are heterozygous for the Christchurch mutation, and these people develop cognitive decline much later than predicted, with fewer vascular amyloid pathologic features and low levels of Tau in the brain. These clinical data strongly support that ApoE3 R136S heterozygote is sufficient to be protective against AD pathology.[50,51] Our results indicate a slight increase in ApoE3ch expression compared with ApoE3 WT (Figure S4J) and comparable mRNA expression between ApoE3 WT and ApoE3ch (Figure S5H). This could be related to half-life or protein stability. Since ApoE3 R136S binds to Tau much stronger than WT, preventing Tau cleavage by AEP (Figure 1). Conceivably, ApoE3 R136S association with Tau may stabilize its protein half-life. Notably, the amyloid and Tau pathologies are quite different from other forms found in AD patients. Longitudinal studies of *in vivo* PET imaging and postmortem neuropathology analyses validated this observation.[6] This discrepancy between Aβ and Tau pathology indicates that putative APOE3ch protective molecular mechanisms may be modulated further by other pathological events during the course and progression of disease. Remarkably, decreased Tau cortical pathology, with the exception of the occipital cortex, which showed the same degree of Tau pathology regardless of age of onset.[44]

Single amino acid substitutions in the N-terminal moiety of ApoE have been shown before to affect the proteins biophysical behavior, function, and role in disease pathogenesis. Specifically, functional differences between ApoE4 and ApoE3 have been attributed to the altered interdomain interactions brought about by the polymorphism at position 112 of the N-terminal domain.[4] Conceivably, R136S mutation may influence the thermodynamic stability and conformation of full-length ApoE3 in a similar manner. In the lipid-free state, ApoE is folded into two independent structural domains with a 22-kDa N-terminal fragment (residues 1–191) and a 10-kDa C-terminal fragment (residues 216–299).[52,53] The protein exists in a predominantly tetrameric form in lipid-free state.[54] Single amino acid variation affects the stability and conformational distribution of ApoE3 in solution, leading to its stronger binding capability toward Tau than WT ApoE3 and protection of the former from AEP cleavage at N368 residue (Figure 1). AEP is activated in neurons in an age-dependent manner, mediating the fragmentation of Tau and APP.[7,8] ApoE protein is primarily synthesized in astrocytes in the CNS system under physiological condition. However, single-cell RNA sequencing (RNA-seq) of different cell types from the human brain confirms physiological expression of ApoE in neurons.[55] Moreover, under pathological condition, it is highly expressed in neurons,[56–60] where ApoE and AEP crosstalk and impact AD pathologies.

The R136S mutation is located in a region of ApoE known to play a role in binding to lipoprotein receptors and heparan sulfate proteoglycans (HSPGs).[61] Previous reports showed that compared

Figure 6. ApoE3 R136S attenuates Tau pathology in Tau P301S mice

AAV-control, AAV-ApoE3 WT, or AAV-ApoE3 R136S was injected into the hippocampus of 3-month-old Tau P301S mice. Cytomegalovirus (CMV) promoter was used in the AAV experiment. The mice were sacrificed 3 months after injection.

(A) Immunohistochemistry showed the expression of AT100 in the brains of Tau P301S mice injected with AAV-control, AAV-ApoE3 WT, or AAV-ApoE3 R136S. Scale bar, 100 μm.

(B) Quantitative comparison of AT100 in the brain slides of Tau P301S mice injected with AAV-control, AAV-ApoE3 WT, or AAV-ApoE3 R136S.

(C) Quantitative comparison of p181-Tau levels, measured via ELISA, in TBS and 2% SDS fractions of brain extracts of Tau P301S mice injected with AAV-control, AAV-ApoE3 WT, or AAV-ApoE3 R136S.

(D and E) Immunofluorescence showing AT8, Iba-1, and NeuN in the hippocampus of Tau P301S mice injected with AAV-control, AAV-ApoE3 WT, or AAV-ApoE3 R136S. Scale bar, 100 μm. Quantitative comparison of AT8, Iba-1, and NeuN was shown in (F)–(H). One-way ANOVA followed by Tukey's post hoc test was performed for the analysis, and data are shown as mean ± SEM. $n = 9$ per group. $^{**}p < 0.01$.

(I and J) AEP and nSMase2 activity in the mouse hippocampus injected with AAV-control, AAV-ApoE3 WT, or AAV-ApoE3 R136S.

(K) Western blot showed the levels of ApoE (soluble and insoluble), p-C/EBP β, AEP, AEP-cleaved UNC5C, and active caspase-3 in the mouse hippocampus injected with AAV-ApoE3 WT or AAV-ApoE3 R136S. The cognitive function of the above mice was evaluated before sacrifice.

(L) The latency to mount platforms during the training of the Morris water maze test.

(M) The percentage of time spent in the target quadrant in the probe test of the Morris water maze.

(N) The swim speed of mice in the Morris water maze test.

(O) The percentage of freezing in the contextual fear test of fear conditioning test.

(P) The percentage of freezing in the cued fear test of fear conditioning test. One-way ANOVA followed by Tukey's post hoc test was performed for the analysis, and data are shown as mean ± SEM. $n = 8$ per group. $^{*}p < 0.05$, $^{**}p < 0.01$.

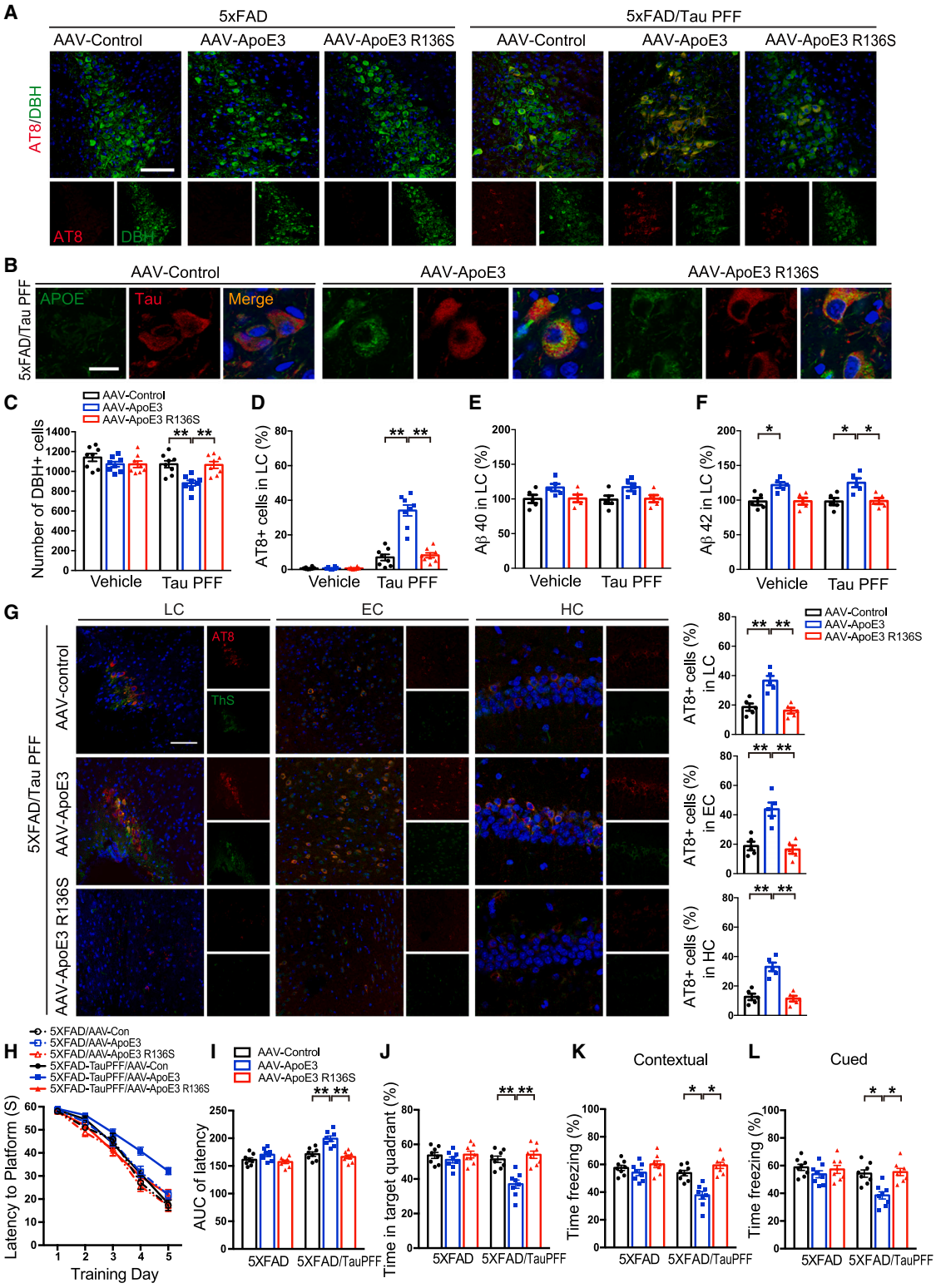


Figure 7. ApoE3 R136S inhibits Tau propagation and LC neurodegeneration in AD mice model

AAV-control, AAV-ApoE3, and AAV-ApoE3 R136S were injected with/without Tau PFFs into the LC of 3-month-old 5xFAD mice. The mice were assessed for Tau phosphorylation, propagation, and memory dysfunction 3 months after injection.

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with ApoE3, ApoE3ch displays a 60% reduction in lipoprotein receptor (LDLR) binding.[18] HSPG promotes Aβ aggregation and neuronal uptake of extracellular Tau, and ApoE-binding may be necessary for some of these effects.[62] In alignment with this idea, aggregation assay in the presence of human recombinant ApoE3 or ApoE3ch proteins, or absence of ApoE showed that Aβ42 aggregation was the highest in the presence of human ApoE3 protein, lower in the presence of human ApoE3ch, and lowest in the absence of any ApoE.[5] Accordingly, expression of ApoE3 R136S in the HC of 5xFAD mice prominently reduces Aβ plaques (Figures S4 and S5). ApoE3 R136S accelerates both the uptake and degradation of Aβ in microglia (Figures 4A–4D). Ultimately, this results in Aβ aggregates reduction in 5xFAD mice (Figure S4). This finding is consistent with the most recent report that ApoE3 R136S reduces Aβ pathology in APP/PS1 mice.[21] Together, these data support that R136S may reduce Aβ accumulation and accelerate Tau degradation (as Tau is reduced) inside microglia. Noticeably, heparin-binding assay showed that ApoE4 has higher affinity for heparin than ApoE3 and ApoE2.[63] Compared with other ApoE isoforms, ApoE3ch displays the lowest heparin-binding ability. Interestingly, a monoclonal antibody (1343A) against ApoE amino acids 130–143 (including R136S) reduces ApoE3 WT and heparin binding, suggesting that modulating ApoE-HSPG interactions could reproduce the protective effect of ApoE3ch.[5]

Single amino acid changes in the functionally important region 136–150 of ApoE3 can affect the molecule's stability and conformation in solution and may underlie functional consequences. Thermal and chemical unfolding analysis revealed small thermodynamic alterations in each variant compared with ApoE3 WT, as well as effects in the reversibility of the unfolding transition.[4] Lipid-bound ApoE3 mutants display reduced binding capacity to the LDL receptor and HSPG compared with ApoE3.[64,65] A previous study showed that Tau pathology spreading is mediated by one of LDL receptors LRP1, which controls the endocytosis of Tau and its subsequent spread. Knockdown of LRP1 significantly reduced Tau uptake in H4 neuroglioma cells and neurons. Downregulation of LRP1 in an *in vivo* mouse model of Tau spread effectively reduces the propagation of Tau between neurons.[17] Here, we show that ApoE3 R136S mutant exhibits stronger binding activity to Tau than ApoE3 WT (Figure 1). Presumably, reduced binding affinity between R136S and LRP1 receptor and HSPG attenuates ApoE3 mutant-bound Tau uptake into neurons or microglia compared with ApoE3 WT, diminishing its endocytosis and spreading between cells (Figures 3 and 4). The relatively low secretion of inflammatory factors and less nSMase2 activity induced by R136S-Aβ complexes may result from less exocytosis (Figures 4I–4L). Ultimately, this leads to a decrease in inflammatory factors and a decrease in exosome secretion with propagating properties, which exerts protective effects on neurons (Figures S3I and S3J). Conceivably, our findings support that the Christchurch variant may ameliorate the distribution of Tau pathology and attenuate neuronal degeneration, leading to slowing down the disease progression and clinical presentation of FAD.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Keqiang Ye (yekeqiang@suat-sz.edu.cn).

Materials availability

This study did not generate new, unique reagents.

Data and code availability

- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

K.Y. conceived the project, designed the experiments, analyzed the data, and wrote the manuscript. S.S.K. and G.C. designed and performed most of the experiments and analyzed the data. M.W. performed parts of the experiments and analyzed the data. E.H.A. performed the Biacore analysis. X.L. conducted genotype and bred the transgenic mice. Z.Z. and D.K.H. assisted with data analysis and interpretation and critically read and wrote the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

(A) Representative images of immunofluorescent staining for Tau AT8 (red), DBH (green), and DAPI (blue) in the LC sections. Scale bars, 100 μm.
(B) Representative images showing colocalization of ApoE (green) and Tau (red) in LC neurons. Scale bars, 20 μm.
(C and D) Quantification of DBH+ cells (C) and AT8+ cells (D).
(E and F) The change of Aβ40 (E) and Aβ42 (F) by ApoE3 or ApoE3 R136S was measured by ELISA. One-way ANOVA was used for the analysis, and data are shown as mean ± SEM. $n = 8$ for (C) and (D), $n = 5$ for (E) and (F). $^{*}p < 0.05$, $^{**}p < 0.01$.
(G) Representative images of immunofluorescent staining for Tau AT8 (red), ThS (green), and DAPI (blue) in the LC, EC, and hippocampus (HC) of virus-injected 5xFAD mice. Quantification of AT8+ cells demonstrated that propagation of AT8+ cells was inhibited by ApoE3 R136S. All scale bars, 100 μm. One-way ANOVA was used for the analysis, and data are shown as mean ± SEM. $n = 5$. $^{**}p < 0.01$.
(H–L) Morris water maze tests demonstrated alleviated memory dysfunctions by ApoE3 R136S mutation in 5xFAD mice treated by Tau PFFs. Shown are the latency traveled to the platform (H), area under the curve for total time traveled (I), and percent time spent in the quadrant platform during the probe trial (J).
(K and L) Fear conditioning tests. Shown are the percentage of freezing time during the contextual fear (K) and cued fear (L) tests following fear conditioning. One-way ANOVA was used for the behavior analysis, and data are shown as mean ± SEM. $n = 8$ per group. $^{*}p < 0.05$, $^{**}p < 0.01$.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-Tau FL (Tau5)	Sigma-Aldrich	cat# MAB361, PRID: AB_94944
Anti-AT8 (Ser202, Thr205)	Invitrogen	cat# MN1020, PRID: AB_223647
Anti-T22 (Oligomeric Tau)	Millipore	cat# ABN454, PRID: AB_2888681
Anti-AT100 (Thr212, Ser214)	ThermoFisher	cat# MN1060, PRID: AB_223652
Anti-Tau (HT7)	ThermoFisher	cat# MN1000, PRID: AB_2314654
Anti-Tau N368	Ye's Lab	N/A
Anti-APOE	ThermoFisher	cat# 701241, PRID: AB_2532438
Anti-β-tubulin	Sigma-Aldrich	cat# T5168, PRID: AB_477579
Anti-β-actin	Abcam	cat# Ab8227, PRID: AB_2305186
Anti-AEP	Cell Signaling	cat# 93627, PRID: AB_2800205
Anti-TSG101	Abcam	cat# Ab30871, PRID: AB_2208084
Anti-UNC5C	Ye's Lab	N/A
Anti-UNC5C N467	Ye's Lab	N/A
Anti-cleaved Caspase3	Cell Signaling	cat# 9661, PRID: AB_2341188
Anti-C/EBPβ	Santa Cruz	cat# SC-150, PRID: AB_2260363
Anti-p C/EBPβ	Cell Signaling	cat# 3084, PRID: AB_22260359
Anti-NeuN	Millipore	cat# MAB377, PRID: AB_2298772
Anti-MAP2	Sigma-Aldrich	cat# M9942, PRID: AB_477256
Anti-Aβ	BioLegend	cat# 800701, PRID: AB_2564633
Anti-IBA1	ThermoFisher	cat# PA5-18039, PRID: AB_10982846
Anti-Akt	Cell Signaling	cat# 4691S, PRID: AB_915783
Anti-pAkt	Cell Signaling	cat# 9271S, PRID: AB_329825
Anti-ERK1/2	Cell Signaling	cat# 9102S, PRID: AB_330744
Anti-pERK1/2	Cell Signaling	cat# 9106S, PRID: AB_331768
Bacterial and virus strains		
AAV-Control	Genecopia	AA001-100
AAV-APOE3 WT	Genecopoeia	AA01-C0189-AV01-200
AAV-APOE3 R136S	Genecopoeia	AA02-CS-C0189-AV01-01-200
AAV-Tau	Emory Vector Core	N/A
Chemicals, peptides, and recombinant proteins		
EMEM	ATCC	cat# 30-2003
DMEM/F12	Gibco	cat# 11320033
Lipofectamine 3000	ThermoFisher	cat# L3000150
HaltTM Protease Inhibitor Cocktail, EDTA-free (100x)	ThermoFisher	cat# 87785
Aβ1-42	rPeptide	cat# A-1163-2
Atto 550 NHS ester	Sigma-Aldrich	cat# 92835
AEP substrate Z-Ala-Ala-Asn-AMC	Bachem	cat# 4033201
Thioflavin S	Sigma-Aldrich	cat# T1892
Critical commercial assays		
site-directed mutagenesis kit	Agilent Technologies	cat# 210519
LDH assay	Promega	cat# G1780
TUNEL In Situ Cell Death Detection Kit	Roche	cat# 11684817910
HisPurTMNi-NTA Spin Purification Kit	ThermoFisher	cat# 88229

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
nSMase2 activity assay	Echelon Biosciences Inc	cat# K-1800
mouse Aβ1-40 ELISA kits	Invitrogen	cat# KMB3481
mouse Aβ1-42 ELISA kits	Invitrogen	cat# KMB3441
human Aβ1-40 ELISA kits	Invitrogen	cat# KHB3481
human Aβ1-42 ELISA kits	Invitrogen	cat# KHB3441
pT181-tau ELISA kits	Invitrogen	cat# KHO0631
IL-1β ELISA kits	ThermoFisher	cat# 88-7013-22
IL-6 ELISA kits	ThermoFisher	cat# 88-7064-22
TNFα ELISA kits	ThermoFisher	cat# 88-7324-22
Proteome Profiler Human Cytokine Array	R&D Systems	cat# ARY005B
Experimental models: Cell lines		
HMC3	ATCC	CRL-3304™
HEK293 cells	ATCC	CRL-1573
SH-SY5Y	ATCC	CRL-2266
Other		
0.22 μm filter Millex	EMD Millipore	cat# SLG5033SS

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

Wild-type mice (all on a C57BL/6J background), P301S mice and 5xFAD were obtained from the Jackson Laboratory (stock numbers: 000664, 008169 and 034840, respectively). 9 female mice per group were used for these studies. ApoE-/- mice on a C57BL/6J background were obtained from Cyagen Biosciences (Jiangsu, China). We crossed ApoE -/- mice with P301S mice to produce P301S/ApoE +/- offspring and then intercrossed these mice to produce P301S /ApoE-/- mice. Investigators were blinded to the group allocation during the animal experiments. Animal care and procedures were conducted according to the National Institutes of Health Guide for Care and Use of Laboratory Animals and approved by the Institutional Animal Care and Use Committee at Emory University.

Transfection and viral infection of cells

Primary cortical neurons were cultured as previously describe (PMID: 25208620). All rats were purchased from the Jackson Laboratory. HMC3 (ATCC, CRL-3304™) is a microglial cell that was isolated from the brain of a patient. HMC3 cells were cultured in advanced EMEM (ATCC, 30-2003)/F12 (Gibco) supplemented with 10% FBS, penicillin (100 U/ml), and streptomycin (100 μg/ml). HEK293 cells (ATCC, CRL-1573) were cultured in high-glucose DMEM with 10% FBS and penicillin (100 U/ml)-streptomycin (100 μg/ml; all from Hyclone). SH-SY5Y (ATCC, CRL-2266) cells were cultured in advanced DMEM/F12 (Gibco) supplemented with 10% FBS, penicillin (100 U/ml), and streptomycin (100 μg/ml). Cells were incubated in a humidified atmosphere of 5% CO_2 at 37 °C. His-hTau, His-hTau 1-368, GFP-Tau, GFP-ApoE3 WT and GFP-Control plasmids were from Ye's Lab. The mutation R136S for ApoE3 was introduced using site-directed mutagenesis kit (Agilent Technologies, 210519), and the mutation was verified by sequencing. HEK293 cells and SH-SY5Y cells were transfected with plasmids encoding Tau, ApoE3 WT, or ApoE3 R136S, using polyethylenimine (PEI) or Lipofectamine 3000 (Thermo Fisher Scientific). The ApoE3 WT and ApoE3 R136S proteins secreted in the conditional medium were collected as described previously for further experiments. AAV-Control (AA001-100, CMV promotor, serotype 2, Genecopia), AAV-APOE3 WT (AA01-C0189-AV01-200, Genecopoeia), AAV-APOE3 R136S (AA02-CS-C0189-AV01-01-200. Genecopia), or AAV-Tau (Emory Vector Core) were used for infection in primary neuron and SH-SY5Y (ATCC® CRL-2266™) neuroblastoma cells. Neurotoxicity was analyzed using LDH assay (CytoTox 96® Non-Radioactive Cytotoxicity Assay, Promega).

METHOD DETAILS

Generation of Aβ PFFs

Recombinant Aβ1-42 HFIP was purchased from rPeptide®. The fibril was prepared as described previously,[66] by adding 900 μl of deionized water to the vial following 2 s vortex and incubation on ice for 5 mins. Repeated the procedure for 30 mins. Then 100 μl of fibril-forming buffer (0.2 M NaPi, 1.5 M NaCl, 0.2% NaN3, pH 7.5) was added to the vial to make final concentration at 1 mg/ml and vortex. Seal tube and place at 37°C for one week, and vortexed the tube daily (15 s each time). Fibrils are stored at −80°C. Fibril formation was validated by Th-T assay.

Protein purification and preparation of hTau fibrils

His-hTau FL, and His-hTau 1-368 were purified from E. coli through Ni-chelating affinity chromatography and eluted with 125 mM imidazole. The purity (>95%) was illustrated by SDS-PAGE gel analysis. The purified hTau protein was induced to aggregate as described previously.[67] Protein fibrillization was verified using the thioflavin S (ThS) fluorescence assay. The hTau fibrils were examined under an electron microscope and Coomassie blue staining.

Uptake and degradation experiments

The preformed Aβ and hTau fibrils (PFFs) were sonicated and labled with Atto 550 NHS ester (Sigma-Aldrich, 92835) according to the manufacturer's instructions. To transfect HMC3 cells, Lipidized ApoE proteins and Atto 550-PFFs (5 μg) were mixed to obtain a 250 μL mixture, incubated for 20 min before being added to the medium. The final concentration of PFFs in the medium was 150 ng/mL.

Regents and antibodies

Regents: Beta-Amyloid (1-42) (rPeptide, A-1163-2), AEP substrate Z-Ala-Ala-Asn-AMC (Bachem, 4033201), the TUNEL In Situ Cell Death Detection Kit (Roche, 11684817910), Halt™ Protease Inhibitor Cocktail, EDTA-free (100x) (Thermo Fisher, 87785), HisPur™Ni-NTA Spin Purification Kit (Thermo Fisher, 88229);

Primary antibodies to the following targets were used: pTau (Ser202-Thr205) (ThermoFisher, MN1020, AT8, 1:1,000 dilution for western blotting and 1:200 for immunofluorescence and immunohistochemistry),Tau (210–241) (Tau 5, Sigma-Aldrich, MAB361, 1:2,000 dilution for western blotting), AT100 (Thermo Fisher Scientific, MN1060, 1:250 for immunohistochemistry), Tau (159-163)(HT7, Thermo Fisher, MN1000, 1:2,000 dilution for western blotting), Tau N368 (Ye Lab, 1:1,000 for western blotting; 1:600 and 1:200 for immunofluorescence), T22 (Millipore, ABN454, 1:1,000 dilution for western blotting and 1:200 for immunofluorescence and immunohistochemistry), ApoE (Thermo Fisher Scientific, 701241, 1:2,000 dilution for western blotting), alpha-tubulin (Sigma-Aldrich, T5168, 1:3,000 dilution for western blotting), AEP (Cell signaling, 93627S, 1:2,000 dilution for western blotting), TSG101 antibody (abcam, ab30871, 1: 500 dilution for western blotting), UNC5C (Ye Lab, 1:2,000 dilution for western blotting), N467 (Ye Lab, 1:1,000 dilution for western blotting), cleaved Caspase 3(Cell signaling technology, #9661, 1:1,000 dilution for western blotting), C/EBPβ (Santa Cruz, sc-150, 1:1,000 dilution for western blotting), p-C/EBPβ (CST, #3084, 1:500 dilution for western blotting), Actin(Abcam, ab8227, 1:3,000 dilution for western blotting), NeuN (MAB377, 1:300 dilution for immunofluorescence), MAP2 (Sigma-Aldrich, M9942, 1:500 dilution for immunofluorescence), Aβ (BioLegend, 800701, 4G8, 1:400 dilution for immunohistochemistry and immunofluorescence), IBA1 (ThermoFisher, PA5–18039, 1:500 dilution for immunofluorescence), AKT (Cell Signaling Technology, 4691s, 1:1,000 dilution for western blotting), pAKTS473 (Cell Signaling Technology, rabbit polyclonal, 9271s, 1:1,000 dilution for western blotting), ERK1/2 (Cell Signaling Technology, 9102s, 1:1,000 dilution for western blotting), pERK1/2 (Cell Signaling Technology, 9106s, 1:2,000 dilution for western blotting).

Purification of extracellular vesicles

The previous methods were followed for isolation of EVs.[68] The conditional medium from HMC3 cells was centrifuged at 2,300g for 30 min to remove cell debris. The supernatant was transferred to a Beckman (Brea, CA, USA) ultracentrifugation tube (#326823) and diluted with 3 ml sterile filtered (0.22 μm filter Millex, EMD Millipore (Billerica, MA, USA) #SLG5033SS) phosphate-buffered saline (PBS) prior to centrifugation for 2.5 h at 120,000g (SW55 rotor K = 48). The supernatant was carefully removed and the pellet was resuspended in lysis buffer for protein quantification, in PBS for IF staining or in Neurobasal medium for functional assays. EVs were isolated from approximately 30 ml of medium (∼25–30 million cells). The primary neurons were treated with the Neurobasal medium containing EVs for 72 h, and were collected for further tests.

Western blot analysis

Cultured cells or mouse brain tissue samples were lysed in lysis buffer (50 mM Tris, pH 7.4, 40 mM NaCl, 1 mM EDTA, 1.5 mM Na_3VO_4, 50 mM NaF, 0.5% Triton X-100, 10 mM sodium pyrophosphate, 10 mM sodium β-glycerophosphate, and a cocktail of protease inhibitors), and centrifuged for 15 min at 15,000 g. The supernatant was boiled in SDS loading buffer. The samples were transferred to a nitrocellulose membrane, after SDS-PAGE. Western blot analysis was carried out with various antibodies. Images have been cropped for presentation.

AEP and nSMase2 activity assay

The assay buffer (20 mM citric acid, 60 mM Na_2HPO_4, 1 mM EDTA, 0.1% CHAPS and 1 mM DTT, pH 6.0) of 100 μl including 100 μM AEP substrate Z-Ala-Ala-Asn-AMC (ABchem) was added to culture cell or mouse brain tissue lysates. AMC released by substrate cleavage was quantified using a fluorescence plate reader at 460 nm for 1 h in kinetic mode. The nSMase2 catalyzes the hydrolysis of sphingomyelin into phosphorylcholine and ceramide, which is one of the lipids critical for exosome formation. The nSMase2 activity assay was tested according the manufacturer's instructions (K-1800, Echelon Biosciences Inc.). The Reaction Mixture and the samples were prepared and added to microtiter plates, which were incubated at 37 °C for 4 hours with shaking. The absorbance at 595 nm was recorded using a microplate reader.

ELISA analysis

Cultured cell, conditioned medium or mouse brain tissue lysates were diluted with the reaction buffer (PBS with 5% BSA, 0.03% Tween-20, and protease inhibitor mixture) and analyzed with mouse Aβ1-40 (Invitrogen, KMB3481), mouse Aβ1-42 (Invitrogen, KMB3441), human β1-40 (Invitrogen, KHB3481), human Aβ1-42 (Invitrogen, KHB3441), pT181-tau (Invitrogen, KHO0631), IL1β (ThermoFisher, 88-7013-22), IL-6 (ThermoFisher, 88-7064-22), and TNFα (ThermoFisher, 88-7324-22) ELISA kits according to the manufacturer's instruction.[8,31] The sample concentrations were determined by comparison with the standard curve.

Cytokine array

Cytokine levels in conditioned medium (CM) were quantified using the Proteome Profiler Human Cytokine Array (R&D Systems; #ARY005B), following the manufacturer's instructions. Pixel density was analyzed using the ImageJ Software.

Biacore X100 binding assay

The activation of the sensor chip was performed by injecting 60 μl of buffer containing 0.2 M N-ethyl-N'–(3-dimethylaminopropyl) carbodiimide and 0.05 M N-hydroxysuccimide into the carboxylated dextran matrix. The activation buffer was prepared from the kit of EDC (1-ethyl-3-(3-demethylamino) propyl carbodiimde) and NHS. We used 3-minutes running time for immobilization step. Purified Tau proteins were immobilized in 10 mM sodium acetate pH 4.0. 50 mM NaOH (BIAcore, GE healthcare). 1 M ethanolamine, pH 8.5. was used to block the remaining binding sites. Purified proteins (100 μg of protein/ml) of ApoE3 WT or ApoE3 R136S were injected for binding to Tau onto the derivatized sensor chip. Regeneration of the senor chip after each analysis cycle was performed by injecting 20 μl of 1.5 M glycine/HCl buffer pH 3.5 and 6 M guanidinium HCl pH 5.5.

Thioflavin S staining

Tau fibrils were stained with Thioflavin S solution (Sigma-Aldrich. Cat#T1892). The brain sections were incubated in 0.25% potassium permanganate solution for 20 min, followed by incubation in bleaching solution containing 2% oxalic acid and 1% potassium metabisulfite for 2 min. Then, the sections were transferred to blocking solution containing 1% sodium hydroxide and 0.9% hydrogen peroxide and incubated for 20 min. The sections were stained with solution of 0.0125% Thioflavin S in 50% ethanol.

Immunostaining

Free-floating mouse brain sections sliced by cryotome were treated with 0.3% H_2O_2 for 10 min. Sections were washed three times in PBS and blocked in 1% BSA, 0.3% Triton X-100, for 1 h, followed by overnight incubation with a tyrosine hydroxylase (DBH) antibody (Invitrogen, PA3-925; 1: 1000), anti-ApoE, or AT-8 antibody (1: 500) at 4°C. Then, the sections were incubated with the matched fluoro-conjugated secondary antibody for 2 h at room temperature, followed by three washes in PBS. The slides were washed three times in PBS and covered with a glass using mounting solution, after DAPI staining for 5 min.

Stereotaxic injection

AAV-APOE3 WT (GeneCopoeia, AA02-C0189-AV01-200, $\geq 5x10^{12}$ GC/ml) or AAV-APOE3 R136S (GeneCopoeia, AA02-CS-C0189-0, $>5x10^{12}$ GC/ml) were injected into the LC region of mice with Tau PFFs. 3-month-old mice (20 - 25 g) of each group were anesthetized with isoflurane (Piramal Healthcare). Meloxicam (5 mg/kg) was injected subcutaneously for analgesia (Loxicom, Norbrook). Unilateral intracerebral injection of virus was performed stereotaxically at the following coordinates: anteroposterior (AP) -5.4 mm and mediolateral (ML) -1.2 mm relative to bregma, and dorsoventral (DV) -3.7 mm from the dural surface for LC; anteroposterior (AP) -1.94 mm and mediolateral (ML) -1.4 mm relative to bregma, and dorsoventral (DV) -2.2 mm from the dural surface for hippocampus. Two microliters of viral suspension were injected into each site using a 10-μl Hamilton syringe with a fixed needle at a rate of 0.25 μl/min. The needle remained in place for 5 min after the viral suspension was completely injected, and then was slowly removed over 2 min. The mice were placed on a heating pad until recovery from the anesthesia.

Behavioral testing

Morris water maze. Mice were trained in a round, water-filled tub (52'' diameter) in an environment with extra maze cues. Each subject was given 4 trials/day for 5 consecutive days (d), with a 15-min intertrial interval. The maximum trial length was 60 s, and if subjects did not reach the platform in the allotted time, they were manually guided to it. Following the 5 d of task acquisition, a probe trial was presented, during which time the platform was removed and the percentage of time spent in the quadrant that previously contained the escape platform during task acquisition was measured over 60 s. All trials were analyzed for latency and swim speed by MazeScan (CleverSys, Inc.). We monitored body weight, visual ability, and swim speed during behavioral test to verify that the deficits did not result from motor dysfunction.

Fear conditioning. The ability to form and retain an association between an aversive experience and environmental cues was tested with a standard fear conditioning paradigm occurring over a period of 3 d. Mice were placed in the fear conditioning apparatus (7″ W, 7″ D X 12″ H, Coulbourn Instruments), composed of Plexiglas with a metal shock grid floor, and allowed to explore the enclosure for 3 min. Following this habituation period, 3 conditioned stimulus (CS)-unconditioned stimulus (US) pairings were presented with a 1-min intertrial interval. The CS consisted of a 20-s 85-db tone, and the US consisted of 2 s of a 0.5-mA foot shock, which was co-terminated with each CS presentation. One min following the last CS-US presentation, mice were returned to their home

cage. On day 2, the mice were presented with a context test, during which subjects were placed in the same chamber used during conditioning on Day 1, and the amount of freezing was recorded via camera and the software provided by Colbourn. No shocks were given during the context test. On day 3, a tone test was presented, during which time subjects were exposed to the CS in a novel compartment. Initially, animals were allowed to explore the novel context for 2 min. Then the 85-db tone was presented for 6 min, and the amount of freezing behavior was recorded.

Stereological cell quantification

Unbiased stereological cell counting was performed using the Stereo Investigator software (MBF Bioscience, Williston, VT) and Olympus microscope. LC sections of 30 μm at every 6th section were stained with TH antibody and analyzed under the randomly placed counting frames (50 × 50 μm) on a counting grid (120 × 120 μm). Optical dissector of 22 μm with 2 μm upper and lower guard zones was used. The total number of TH+ neurons in the LC was estimated using the optical fractionator method. TH-positive axon terminals in the striatum were estimated using fluorescence intensity by Image J software. For quantification of positive cells and image analysis, stained colour was selected and set the proper threshold for the binarization of the selected colour image by Image J software. The total number and intensity of immune-reactive neurons were analysed using the same threshold. The conditions of the analysis were blinded to the investigator.

QUANTIFICATION AND STATISTICAL ANALYSIS

All cell culture data are expressed as mean ± S.E.M. from 3 or more independent experiments, and the level of significance between 2 groups was assessed with Student's t-test. For experiments consisting of more than 2 groups, one-way or two-way ANOVA followed by Tukey's post-hoc test was applied. A value of $p < 0.05$ was considered to be statistically significant. Sample sizes were determined by Power and Precision (Biostat).